1. Semantic values
2. Models
3. Satisfiability
4. Semantic consequence
5. Validity (Tautology), Contradiction

Logic in Computer Science Lab
No. 4
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QUIZ 1: NEXT WEEK

George Boole
(1815–1864)

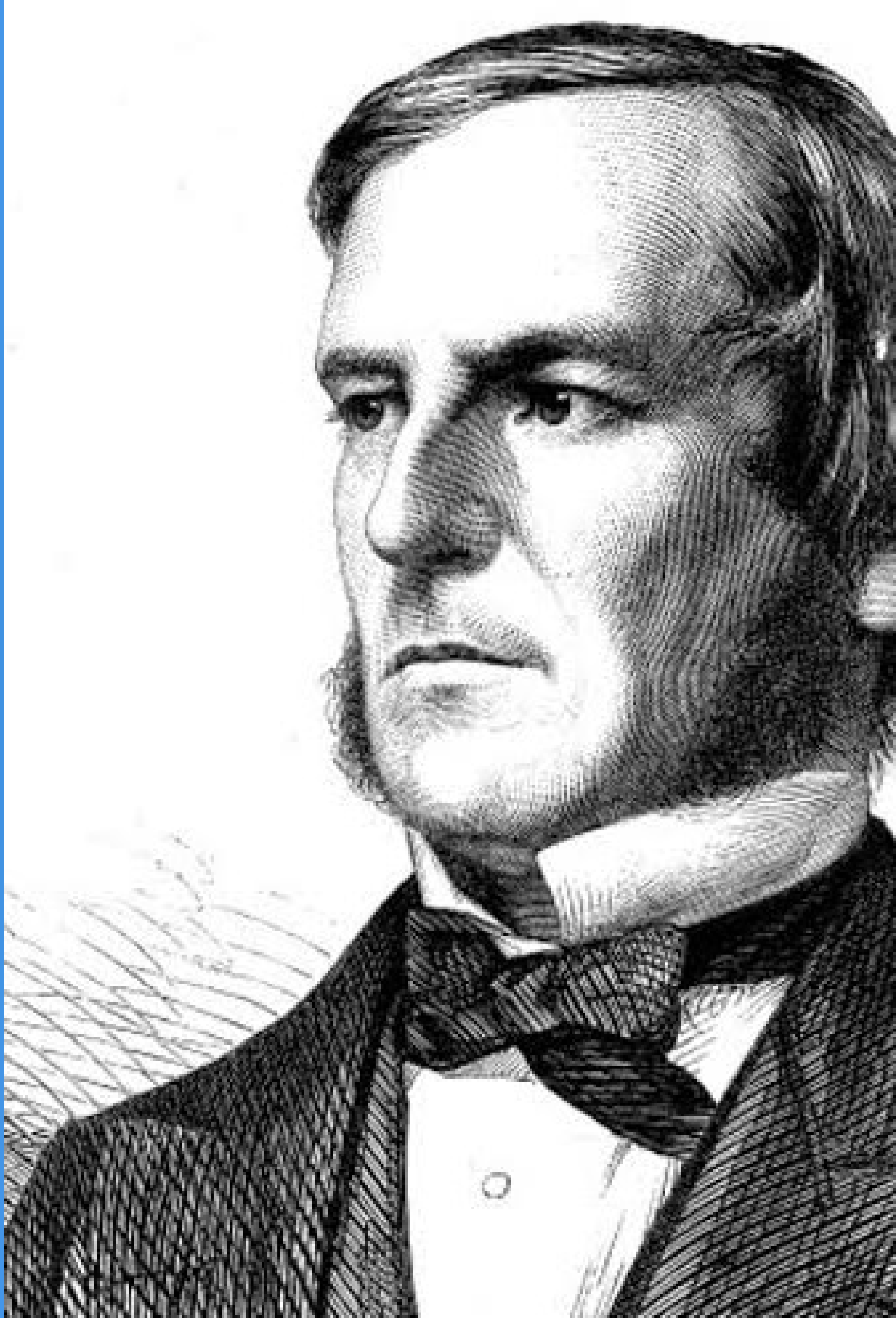
English autodidact, mathematician, philosopher and logician who served as the first professor of mathematics at Queen's College, Cork in Ireland. Creator of Boolean algebra (later renamed). Boolean logic is essential to computer programming.

Shannon (70 years later) demonstrated how Boolean algebra could be used to optimize the design of electromechanical relay systems.

He proved that Boolean algebra problems could also be solved with these circuits.

The use of the properties of electrical switches for logical operations is the basis of all modern computing.

Thus, Boole, with the help of Shannon, created the theoretical foundations of the Digital Age.



Truth table

Truth value or semantic value of a formula

The *truth value of a formula* $\mathcal{F}$ with respect to a given interpretation ϱ is denoted by $|\mathcal{F}|_\varrho$ and is defined recursively as follows:

- if $A \in Con$ then $|A|_\varrho \rightleftharpoons \varrho(A)$,
- if $B \in Form$ and $C \in Form$ then

$$
\begin{aligned}
|\neg B|_\varrho &\rightleftharpoons 1 - |B|_\varrho \\
|(B \wedge C)|_\varrho &\rightleftharpoons \min(|B|_\varrho, |C|_\varrho) \\
|(B \vee C)|_\varrho &\rightleftharpoons \max(|B|_\varrho, |C|_\varrho) \\
|(B \supset C)|_\varrho &\rightleftharpoons \max(1 - |B|_\varrho, |C|_\varrho) \\
|(B \equiv C)|_\varrho &\rightleftharpoons \begin{cases} 1 & \text{if } |B|_\varrho = |C|_\varrho, \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}
$$

2-argument Boolean functions: $f : \{0,1\}^2 \to \{0,1\}$

2-argument dependency

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\Vert$	$\wedge$	$\vert$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

$\oplus$ **XOR**

$\downarrow$ **Peirce's arrow**

$\uparrow$ **Sheffer stroke**

1-argument dependency

Identity 1	Negation 1	Identity 2	Negation 2
π_1	π_1'	π_2	π_2'
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0

No argument dependency

Falsity	Truth
$\perp$	$\top$
0	1
0	1
0	1
0	1

Law of the double negation:

$$\neg\neg A \Leftrightarrow A$$

Truth table – Sample

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\parallel$	$\wedge$	$\mid$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

Give the truth table of the formula $(\neg(p \supset q) \equiv (p \wedge \neg q))$!

p	q	(	$\neg$	$(p$	$\supset$	$q)$	$\equiv$	$(p$	$\wedge$	$\neg$	$q))$
0	0		0		1		1		0	1	
0	1		0		1		1		0	0	
1	0		1		0		1		1	1	
1	1		0		1		1		0	0	

For example, the semantic value of the formula is 1 at the interpretation where $\varrho(p) = 1$ and $\varrho(q) = 1$.

The semantic value of this formula is 1 at every interpretation.

		Equivalent $\equiv$	Not equivalent $\not\equiv$	Implies $\supset$	OR Disjunction $\vee$	NOR $\|$	AND Conjunction $\wedge$	NAND $\mid$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

Exercises

Truth tables

28. Give the truth table of the formula $((p \wedge (q \wedge r)) \equiv ((p \wedge q) \wedge r))$!

29. Give the truth table of the formula $((p \vee (q \vee r)) \equiv ((p \vee q) \vee r))$!

30. Give the truth table of the formula $((p \wedge q) \equiv (q \wedge p))$!

31. Give the truth table of the formula $((p \vee q) \equiv (q \vee p))$!

32. Give the truth table of the formula $((p \wedge (q \vee r)) \equiv ((p \wedge q) \vee (p \wedge r)))$!

33. Give the truth table of the formula $((p \vee (q \wedge r)) \equiv ((p \vee q) \wedge (p \vee r)))$!

Exercises

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\parallel$	$\wedge$	$\mid$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

44. Give the truth table of the formula $((p \wedge q) \equiv \neg(\neg p \vee \neg q))$!

45. Give the truth table of the formula $((p \vee q) \equiv \neg(\neg p \wedge \neg q))$!

46. Give the truth table of the formula $((p \supset q) \equiv \neg(p \wedge \neg q))$!

47. Give the truth table of the formula $((p \supset q) \equiv (\neg p \vee q))$!

48. Give the truth table of the formula $((p \wedge q) \equiv \neg(p \supset \neg q))$!

49. Give the truth table of the formula $((p \vee q) \equiv (\neg p \supset q))$!

Models, satisfiability, validity, logical equivalence

Model

Let $\Gamma \subseteq Form$ be an arbitrary set of formulae of a zero-order language $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$.

- The interpretation ϱ is the *model* of the set of fromulae Γ if $|A|_\varrho = 1$ for all $A \in \Gamma$.

Let $A \in Form$ be a formula fo a zero-order language $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$.

- The interpretation ϱ is the *model* of the formula A if the ϱ is the model of the singleton set $\{A\}$.

Model – Example

p	q	$((p \wedge \neg p) \vee q)$
0	0	0
0	1	1
1	0	0
1	1	1

The formula is true at 2 interpretations, so this formula has 2 models.

Satisfiability – Unsatisfiability

Let $\Gamma \subseteq Form$ be an arbitrary set of formulae of a $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ zero-order language.

- The Γ set of formulae is *satisfiable*, if it has a model.
- The Γ set of formulae is *unsatisfiable*, if it is not satisfiable.

Let $A \in Form$ be an arbitrary formula of a $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ zero-order language.

- The formula A is *satisfiable* if the $\{A\}$ singleton set is satisfiable.
- The formula A is *unsatisfiable* if the $\{A\}$ singleton set is unsatisfiable.

Satisfiability, Unsatisfiability – Example

p	q	$((p \land \neg p) \lor q)$
0	0	0
0	1	1
1	0	0
1	1	1

The formula has at least 1 model, so this formula is

- satisfiable
- not unsatisfiable

Validity, Tautology – Example

Give the truth table of the formula $(\neg(p \supset q) \equiv (p \wedge \neg q))$!

p	q	$($	$\neg$	$(p$	$\supset$	$q)$	$\equiv$	$(p$	$\wedge$	$\neg$	$q))$
0	0		0		1		1		0	1	
0	1		0		1		1		0	0	
1	0		1		0		1		1	1	
1	1		0		1		1		0	0	

The semantic value of this formula is 1 at every interpretation, that is, this formula is valid.

Tautology – Contingency – Contradiction

Contingent propositions are

neither true under every possible valuation (i.e., tautologies)

nor false under every possible valuation (i.e., contradictions)

p	q	$((p \land \neg p) \lor q)$
0	0	0
0	1	1
1	0	0
1	1	1

Contingent formula

Satisfiable

Tautology – Contingency – Contradiction

Valid formula Tautology

p	$p \vee \neg p$
0	1
1	1

Satisfiable

Model 1: $|p|_\varrho = 0$

Model 2: $|p|_\varrho = 1$

Contingent formula

p	q	$((p \wedge \neg p) \vee q)$
0	0	0
0	1	1
1	0	0
1	1	1

Satisfiable

Model 1: $|p|_\varrho = 0$ and $|q|_\varrho = 1$

Model 2: $|p|_\varrho = 1$ and $|q|_\varrho = 1$

Contradiction

p	$p \wedge \neg p$
0	0
1	0

Unsatisfiable / Not satisfiable

It has no model

Semantic consequence

Let $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language, $\Gamma \subseteq Form$ be an arbitrary set of formulae, and $A \in Form$ be an arbitrary formula of the language.

- The formula A is the *semantic consequence* of Γ if $\Gamma \cup \{\neg A\}$ is unsatisfiable.
  (Notation: $\Gamma \models A$)

Semantic consequence

Is the formula q a semantic consequence of the two formulae p and $(p \supset q)$?

p	q	p	$(p \supset q)$	q
0	0	0	1	0
0	1	0	1	1
1	0	1	0	0
1	1	1	1	1

The two formulas (hypotheses) are true at one interpretation only (in the highlighted line), and in this interpretation the formula q (the conclusion) is also true.

It means that the conclusion (the formula q) is a logical consequence of the 2 hypotheses.

Valid formulas, Tautology

Let $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language, $\Gamma \subseteq Form$ be an arbitrary set of formulae, and $A \in Form$ be an arbitrary formula of the language.

- The formula A is *valid* (*tautology*) if
  $\boxed{\emptyset \models A.}$
  (Notation: $\models A$)

Logical equivalence

Let $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language and $A \in Form$ and $B \in Form$ be two formulas of $\mathcal{L}^{(0)}$.

The formula A and formula B are *logically equivalent*, if

$$\boxed{A \models B \quad \text{and} \quad B \models A.}$$

(Notation: $A \Leftrightarrow B$)

Logical equivalence – Example

We have two formulae: $(\neg p \vee q)$ and $(p \supset q)$

Are they logically equivalent?

p	q	$(\neg p \vee q)$	$(p \supset q)$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

The two formulae give the same value in every interpretation, so the two formulae are logically equivalent.

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\|\|$	$\wedge$	$\|$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

Exercises

Which of the following sets of formulae are satisfiable or not satisfiable?

If a set is satisfiable, then give a model of this set!

a) $\{(p \wedge \neg q),\ (\neg p \vee \neg q),\ (r \supset \neg p)\}$

b) $\{(\neg p \wedge \neg q),\ (p \vee q)\}$

c) $\{(p \equiv r),\ (q \equiv \neg r),\ (p \vee q)\}$

Exercises

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\Vert$	$\wedge$	$\vert$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

Decide whether the formulae are valid, unsatisfiable, or satisfiable but not valid.

a) p

b) $(((p \supset q) \supset p) \supset q)$

c) $((\neg p \wedge \neg q) \equiv (p \vee q))$

d) $(((p \supset q) \supset r) \supset (p \supset (q \supset r)))$

e) $((p \supset (q \supset r)) \supset ((p \supset q) \supset r))$

Exercises

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\|$	$\wedge$	$\mid$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

2. Is the logical consequence of formulae p and $(p \supset q)$ the formula q? Justify your answer!
3. Is the logical consequence of formulae $(p \supset q)$ and $(q \supset r)$ the formula $(p \supset r)$? Justify your answer!
4. Is the logical consequence of formulae $(p \vee (q \vee r))$, $\neg q$ and $\neg r$ the formula p? Justify your answer!
5. Is the logical consequence of formulae $(p \supset q)$ and $\neg q$ the formula $\neg p$? Justify your answer!
6. Is the logical consequence of formulae $(p \supset (q \supset r))$ and q the formula r? Justify your answer!

Exercises

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\Vert$	$\wedge$	$\vert$
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

12. Which formulae are logical consequences of the formula $(q \supset q)$?
☐ $((q \vee p) \vee p)$ ☐ $(q \vee (p \equiv p))$ ☐ $(q \equiv (p \vee q))$ ☐ $(\neg p \vee \neg q)$ ☐ $((p \vee q) \supset p)$

13. Which formulae are logical consequences of the formula $(p \vee \neg q)$?
☐ $\neg(p \wedge \neg q)$ ☐ $(p \equiv (p \vee q))$ ☐ $((p \wedge p) \vee q)$ ☐ $(p \equiv (q \vee q))$ ☐ $(q \supset (p \supset p))$

14. Which formulae are logical consequences of the formula $((q \equiv q) \equiv p)$?
☐ $\neg(\neg q \vee p)$ ☐ $(p \vee (p \wedge q))$ ☐ $(\neg p \supset \neg q)$ ☐ $(\neg q \supset p)$ ☐ $\neg(\neg p \vee q)$

15. Which formulae are logical consequences of the formula $((p \wedge q) \wedge (r \vee \neg s))$?
☐ $(p \wedge q)$ ☐ $(p \vee \neg s)$ ☐ $(p \wedge \neg s)$ ☐ $(q \supset r)$ ☐ $(s \supset r)$

Exercises

32. Which of the following formulae are logically equivalent to formula $((p \land q) \supset r)$?
    $\square$ $(r \lor \neg(p \land q))$ $\square$ $(p \land (\neg\neg q \land \neg r))$ $\square$ $(p \supset (q \supset r))$ $\square$ $((r \lor \neg p) \lor \neg q)$

33. The formulae $(p \supset (q \supset r))$ and $((p \supset q) \supset r)$ are logically equivalent? Justify your answer!

34. The formulae $((p \land q) \lor r)$ and $(p \land (q \lor r))$ are logically equivalent? Justify your answer!

35. The formulae $((p \supset q) \supset r)$ and $((p \supset r) \supset q)$ are logically equivalent? Justify your answer!

Exercises

62. Which of the following formulae are logically equivalent to formula $(q \vee (p \wedge p))$?
$\square$ $((q \equiv p) \supset p)$ $\square$ $(q \supset (q \supset p))$ $\square$ $((q \supset p) \equiv p)$ $\square$ $((q \supset p) \supset p)$ $\square$ $(\neg q \supset p)$

63. Which of the following formulae are logically equivalent to formula $(q \supset q)$?
$\square$ $(p \vee \neg p)$ $\square$ $(q \equiv (p \supset q))$ $\square$ $(\neg p \supset \neg p)$ $\square$ $(p \supset (q \supset p))$ $\square$ $(q \vee (q \supset q))$

64. Which of the following formulae are logically equivalent to formula $(p \equiv \neg p)$?
$\square$ $(\neg p \wedge q)$ $\square$ $\neg(\neg p \supset p)$ $\square$ $((p \wedge q) \supset q)$ $\square$ $\neg(\neg q \supset p)$ $\square$ $\neg(q \vee \neg q)$

65. Which of the following formulae are logically equivalent to formula $\neg(q \vee p)$?
$\square$ $\neg(\neg p \vee p)$ $\square$ $\neg(q \supset p)$ $\square$ $\neg(p \supset q)$ $\square$ $((p \supset q) \equiv p)$ $\square$ $(p \supset \neg p)$